Customer Behavior Analysis

Alfido Tech Internship – Task 1

Executive Summary

Objective

The objective of this project was to analyze customer purchasing behavior using a real-world dataset to identify customer segments, purchasing patterns, and churn risks. The analysis aims to provide data-driven insights that can help Alfido Tech improve customer engagement, increase customer retention, and support informed business decisions.

Dataset Information

- **Dataset:** Customer Behavior Analysis
- **Source:** Kaggle
- **Tools Used:** Python, Pandas, NumPy, Matplotlib, Jupyter Notebook
- **Analysis Performed:** Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Customer Segmentation (K-Means Clustering), Data Visualization

Data Cleaning & Preparation

- Checked and handled missing values.
- Removed duplicate records.
- Renamed column names for consistency.
- Converted the purchase date into datetime format.
- Created additional features such as Month and Year for trend analysis.

Key Findings

- Customers were segmented into three distinct groups using K-Means clustering based on purchasing behavior.
- Product categories showed different levels of customer demand and revenue contribution.
- Purchase trends varied across different months, indicating seasonal buying behavior.
- Customer churn was observed among a portion of customers, highlighting opportunities for retention strategies.
- Customers preferred specific payment methods, providing insights for improving the payment experience.

Visualizations

(Insert screenshots from your notebook here.)

- Sales by Product Category
- Monthly Purchase Trend
- Customer Segmentation (K-Means Clusters)

Recommendations

1. Introduce a loyalty rewards program for high-value customers to encourage repeat purchases.
2. Launch personalized marketing campaigns for different customer segments.
3. Identify and engage customers at risk of churn with targeted retention offers.
4. Increase promotions for high-performing product categories during peak purchase periods.
5. Improve the checkout experience by supporting customers' preferred payment methods.

Conclusion

The customer behavior analysis provided valuable insights into purchasing patterns, customer segmentation, and churn risk. These findings can help Alfido Tech develop targeted marketing strategies, improve customer satisfaction, increase retention, and drive sustainable business growth through data-driven decision-making.